

# Simulating the Effect of Angle of Attack, Camber, and Thickness on Lift Using a Panel Method

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## I. Nomenclature

|            |   |                                                     |
|------------|---|-----------------------------------------------------|
| $C_L$      | = | lift coefficient                                    |
| $i$        | = |                                                     |
| $I_{ij}$   | = | source panel normal velocity geometric integral     |
| $j$        | = | index referring to the current panel                |
| $J_{ij}$   | = | source panel tangential velocity geometric integral |
| $K_{ij}$   | = | vortex panel normal velocity geometric integral     |
| $L_{ij}$   | = | vortex panel tangential velocity geometric integral |
| $n$        | = | normal vector of a boundary                         |
| $r$        | = | radius                                              |
| $s$        | = | position along a curve                              |
| $u$        | = | x-component of flow velocity                        |
| $v$        | = | y-component of flow velocity                        |
| $V$        | = | velocity                                            |
| $V_n$      | = | velocity component in the normal direction          |
| $V_r$      | = | velocity component in the radial direction          |
| $V_t$      | = | velocity component in the tangential direction      |
| $V_\theta$ | = | velocity component in the angular direction         |
| $V_\infty$ | = | free stream velocity                                |
| $\alpha$   | = | angle of attack                                     |
| $\gamma$   | = | local circulation                                   |
| $\Gamma$   | = | circulation                                         |
| $\lambda$  | = | local source strength                               |
| $\Lambda$  | = | strength of source                                  |
| $\phi$     | = | potential function                                  |
| $\phi_v$   | = | potential function for vortex panel                 |
| $\psi$     | = | stream function                                     |
| $\phi_s$   | = | potential function for source panel                 |
| $\theta$   | = | angle                                               |

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## II. Introduction

The mission of this report was to identify key design parameters for a glider and explore how these characteristics can be adapted to achieve maximum glider performance. Learning and understanding the makings of a glider is key to the study of aerodynamics and aerospace engineering in general. Research into gliders included an exploration of the history of gliders and early flight to reveal lessons learned in early aircraft development. A trade study was also conducted to analyze how gliders have been designed in a more modern context. The investigation revealed that airfoil design, along with aspect ratio, wing configuration, wingtips, and many more variables influence the performance of the wings, quantified as the glide ratio for a glider. In the academic problem, important characteristics of airfoils such as lift coefficient and pressure distribution were analyzed through the use of a combined source and vortex panel method coded in Python. The panel method is a fluid mechanics technique that was discussed thoroughly in lecture. It involves important concepts such as potential flow, vortices, source sheets, vortex sheets, and airfoils. These topics are integral to understanding incompressible flow, and coding a panel method forced the team to grapple with the theory and process involved to successfully execute the panel method.

## III. Real-World Problem

### A. Historical Context

Long before any gliders or heavier-than-air manned flight, and even long before the invention of a wind tunnel, in the mid 18th century, Benjamin Robins invented one of the first aerodynamic test devices. This device, called the whirling arm, was the first to demonstrate that the area of an object's profile seen by the movement through air was not the only factor which determined the drag experienced by the object, but its shape and orientation played a more important role [1]. He also observed that the magnitudes of aerodynamic forces on the body were directly proportional to the square of the velocity of the body, confirming earlier mathematical theories and the effect of the aspect ratio on the aerodynamic forces experienced by the body. The aspect ratio is defined as the ratio of span squared to the wing area, which reduces to the ratio of the span to the chord in the case of a rectangular wing [2]. The specifics of airfoil nomenclature will be described below. Just a few decades later in 1804, Sir George Cayley built the first modern configuration aircraft based on data collected from the whirling arm. Cayley was the first to recognize a bifurcation between the forward motion through the air and a lift force which could support the aircraft; he thought that an aircraft can use some forward propulsion method to push through the air, and the surface of the craft itself could be used to generate lift [3].



**Fig. 1 Sir George Cayley's sketch of the first modern configuration aircraft [3]**

The whirling arm had significant shortcomings, however, because the arm would generate a rotary motion in the surrounding air, producing turbulence which confounded data collection, especially as the speed of the arm was increased. This also made the setup and construction more difficult due to the strong forces involved at high speeds.

In 1871, Frank H Wendham developed the first wind tunnel, using a fan-blower to produce a uniform flow through a 12 foot long chamber with 18 square inches of area. He used various shapes and tested the relationship between the lift to drag ratio and the angle of attack, along with the aspect ratio of the wing, confirming a natural observation of birds: long, narrow wings provide more lift than shorter wings of equal area. Some questioned the extent to which this admittedly small wind tunnel could be reasonably extrapolated and applied to larger full-sized aircraft, but Osborne Reynolds demonstrated that the airflow pattern over a smaller model could be used to approximate the flow over a full sized craft, so long as a constant parameter, called the Reynolds number, was maintained. The Wright brothers made a wind tunnel themselves when the aerodynamic data they were using at the time failed to yield results [1].



**Fig. 2 The wind tunnel constructed by the Wright brothers in 1901 [4]**

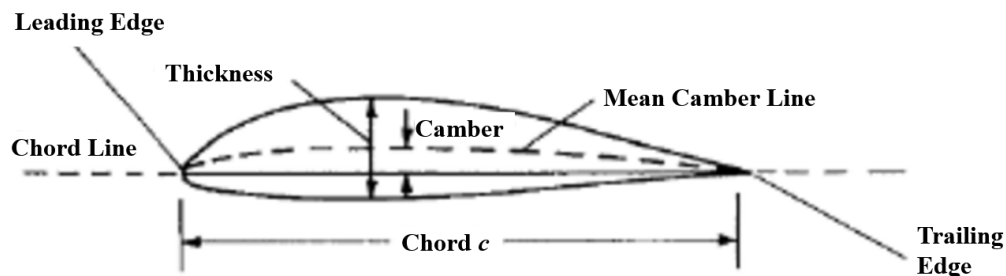
## **B. Airfoils**

An airfoil is simply a 2 dimensional cross section of a wing, conceptualized to infinitely extend into and out of the page, constituting an ‘infinite wing’. At the tail end of the first world war, Ludwig Prandtl showed that analysis of an infinite wing could be used to make calculations for a finite wing [5]. This analysis is conducted by making a few simplifications and building a mathematical representation of the wing. In the academic problem discussed below, these assumptions and mathematical derivations will be made in order to make predictions and compare the performance of airfoils. From this model, the lift per unit span can be calculated, the coefficient of lift and coefficient of moment about any point can be found, and the relationship between angle of attack and coefficient of lift can be observed. These models depend on the shape of the airfoil and some key characteristics, which will briefly be covered in this report, but do not correctly predict drag. The consequence of d’Alembert’s paradox is that the drag on an airfoil will be zero, which is not accurate or realistic, but the lift and moment on an airfoil result from the pressure distribution, which is correctly predicted by inviscid flow [5].

### *1. Airfoil Nomenclature*

The measurements of specific characteristics of an airfoil affect the performance of the airfoil [6].

- Mean Camber Line: the locus of points halfway between the upper and lower surfaces as measured perpendicular to the mean camber line itself
- Leading Edge: the most forward point of the mean camber line
- Trailing Edge: the most rearward point of the mean camber line
- Chord line: the straight line connecting the leading and trailing edges.
- Chord  $c$ : the distance from the leading edge to the trailing edge; the length of the chord line
- Camber: the maximum distance between the mean camber line and the chord line, measured perpendicular to the chord line
- Thickness: the distance between the upper and lower surfaces, measured perpendicular to the chord line



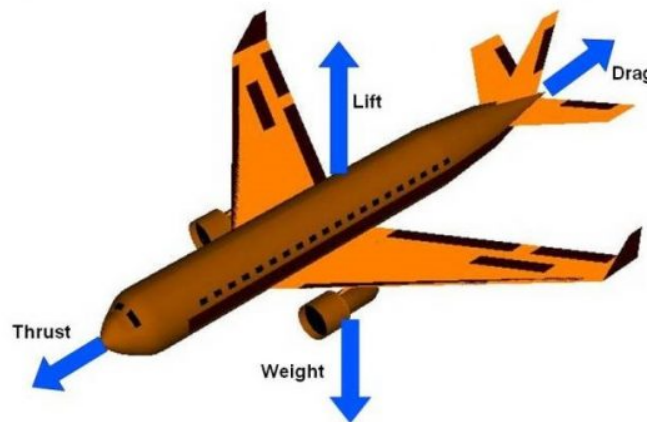
**Fig. 3 Airfoil diagram with key characteristics labeled [311 book]**

Otto Lilienthal was the first to formally define modern aerodynamic force coefficients to make measurements on cambered airfoils, despite still using a whirling arm. Lilienthal was also the first to make drag polar plots which

illustrated the effect of angle of attack on lift and drag for a given airfoil [5]. The limitations of small wind tunnels and whirling arm gave early aerodynamicists, including Cayley, Lilienthal [7], and later even Wright and Langley, the false impression that extremely thin airfoils were optimal. This misconception led to airfoils so thin that flow separation occurred at low angles of attack, and therefore they had low maximum lift coefficients. The first man to identify thicker airfoils as a viable alternative was Prandtl, who discovered that increased thickness greatly increased the maximum lift coefficient before stall and provided greater structural integrity and design flexibility, since an internal cantilever design could be used instead of the wire bracing used in early aircraft [5].

### C. Lift and Drag

The forces experienced by an aircraft in flight are Lift, Drag, Thrust, and Weight. The weight is simply the gravitational force on the aircraft, and the thrust is the forward force on the aircraft by its propellers or engine (nothing in the case of a glider) [8][9]. The amount of lift generated by an aircraft results primarily from the pressure difference between the lower surface of the wings and the upper surface of the wings. The Kutta-Joukowski theorem also states that the circulation of air around an airfoil is also directly related to the lift per unit span of the airfoil:  $L' = \rho_{\infty} V_{\infty} \Gamma$  [5]. This pressure distribution and/or circulation can be calculated to determine the lift per unit span generated by an airfoil or coefficient of lift, which will vary with angle of attack. The drag experienced by an aircraft depends primarily on the shape, size, and speed of the craft, as well as the altitude. This calculation can be estimated for an aircraft, and experimental testing can determine the drag coefficient for an airfoil, but inviscid potential flow cannot be used to accurately determine drag. This is due to skin friction and flow separation, since these are viscous effects and depend on Reynolds number.



**Fig. 4 The four forces on an airplane.[8]**

The lift and drag coefficients are used to calculate lift and drag and are directly proportional to them for the same conditions. The relationship between the angle of attack and the lift coefficient can be plotted using observations made from wind tunnels and inviscid flow on airfoils. Experimental analysis of an airfoil tested in a wind tunnel will still give the most accurate maximum lift coefficient for an airfoil, but the panel method used below can still reliably give the lift coefficient as long as there is no flow separation. Flow separation occurs when the flow at the boundary layer closest to the airfoil separates from the surface of the airfoil, causing a wake effect and drastically increasing drag. This condition is called stall. Therefore, it is desirable to achieve a high maximum lift coefficient to avoid stall. The angle of attack at which the airfoil achieves a lift coefficient of zero is the zero-lift angle of attack. This point, along with the slope of the lift to angle of attack curve are the primary things inviscid potential flow theory is able to reasonably predict [5].

### D. Trade Study

Now a trade analysis of gliders and airplanes in history and industry will guide the search for important glider characteristics. Not every glider or plane will have the same design goals or available measurements, so this exploration will be somewhat open ended, with the results interpreted in the following section. The goal of the study is to identify commonalities between gliders, including both characteristics and requirements.

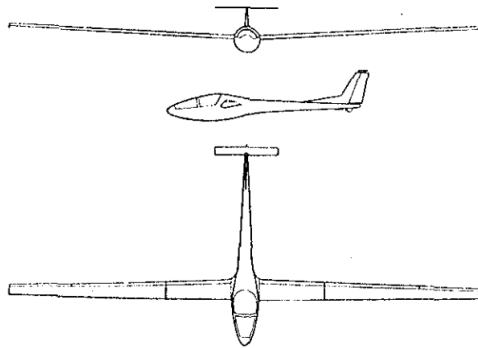
One glider – developed in 2019 for DARPA to supply combatants in hostile conditions – is the LG-2K RAIN

glider [10]. This glider is intended as a disposable supply drop to be deployed in combat zones in a variety of weather conditions. The RAIN is deployed from a cargo plane in flight at 130 knots of airspeed, and carries up to 1,600lb of cargo a distance of over 70 nautical miles, with an intended glide ratio of up to 15 to 1. The glider's wingspan is 23.7 feet. The glider uses a symmetric airfoil primarily for ease of construction, since its wings are retractable and thus have less room for additional supports, and also because a single support at the quarter chord of the aircraft will have a zero-pitching moment for a symmetric airfoil.



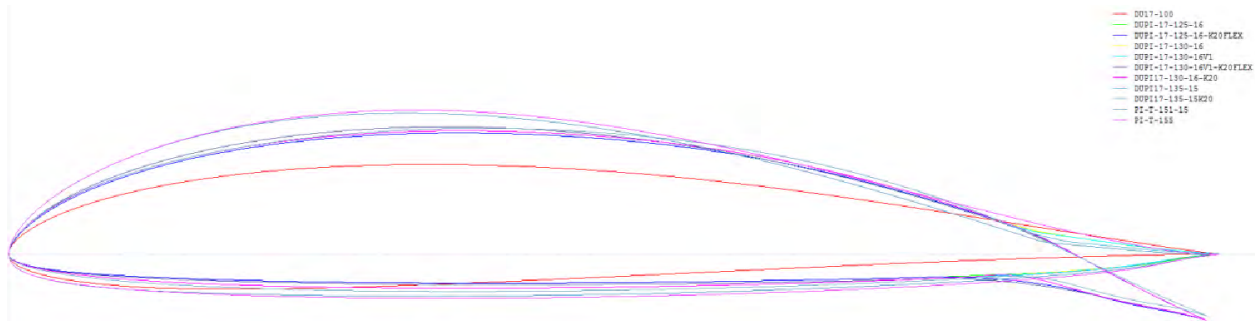
**Fig. 5 LG-2K RAIN glider in flight [10]**

In 1974, D.J. Marsden from the University of Alberta developed a variable geometry sailplane called the Gemini. A sailplane is a piloted glider built with standard aircraft parts that does not have an engine [9]. The Gemini was built to perform better than current gliders in the open class of racing gliders and to have good handling both in flight and on the ground. The sailplane is designed for recreational purposes. The Gemini has a wingspan of 60.5 feet, an aspect ratio of 29.4, a best glide ratio of 40:1 at 56 knots, and a stall speed of 35 knots [11]. To achieve the first objective, Marsden chose a high aspect ratio and a large wingspan. To achieve the second, he included a full span slotted flap for maneuvering at low speeds. The subjective reported experience is that the plane performs excellently.



**Fig. 6 Three view line drawing of the Gemini [11]**

Another sailplane developed in a more modern context by Poulo Iscold, Jim Payne, and Loek M.M. Boermans is Nixus. This sailplane is the first fly-by-wire (FBW) sailplane, allowing for automated stabilization using flaps. The plane's design objectives are to achieve "maximum performance" as a sailplane while using this novel technology [12]. In a normal plane, the pilot must make many small adjustments to maintain the maximum glide ratio. The FBW system is intended to automate these adjustments and achieve greater efficiency compared to a pilot, putting less stress on the pilot. Some other advantages include an ability to utilize higher aspect ratio wings and thinner airfoils and avoiding high control forces. The authors also mention control over the mixing ratio for ailerons and flaps, and further claim that future development will lead to load alleviation and flutter suppression. Potential dangers identified are a need for electrical power, safety modes, and anti-symmetry devices or strategies. Nixus has a wingspan of 92 feet and an aspect ratio of 53.3 [12].



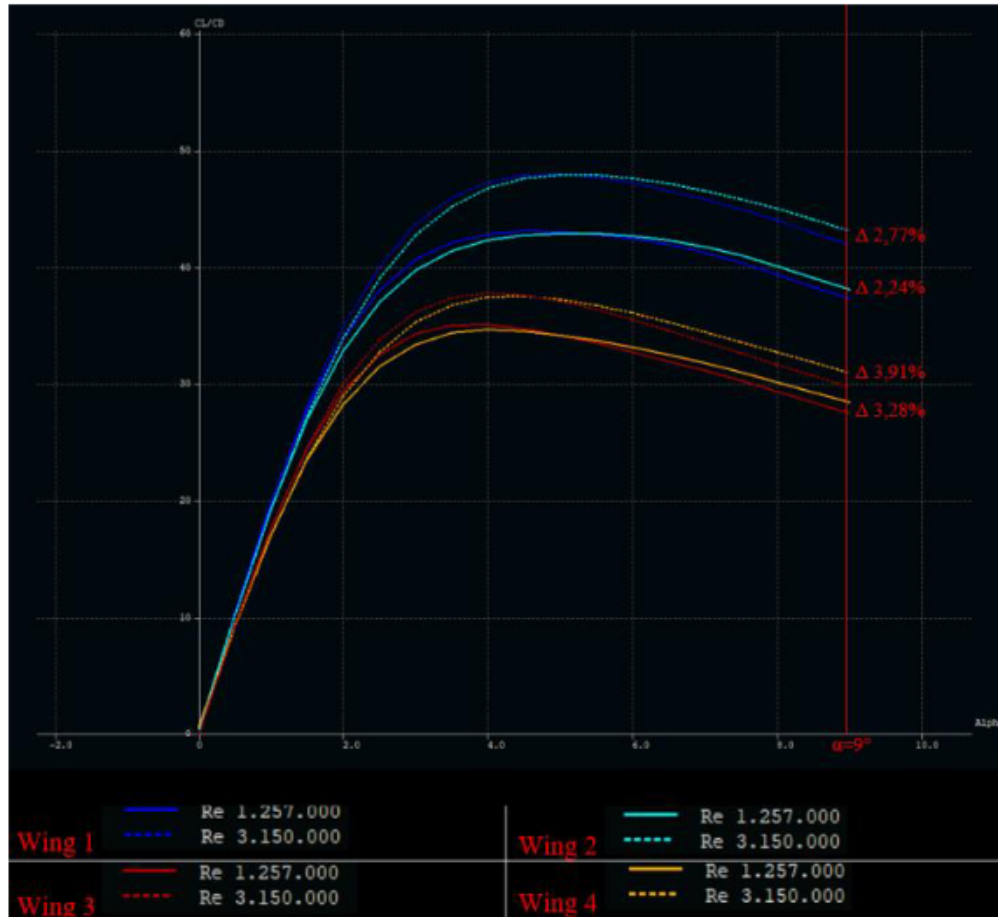
**Fig. 7** An overlay of the 11 airfoils present along the span of one of Nixus's wings [12]

### E. Key Characteristics and Additional Considerations

The historical and trade analyses conducted have painted a clear picture of the important factors when building a glider. There are also additional considerations and evidence that must be addressed before making any decisions. One critical metric is the glide ratio [13]. The glide ratio is the ratio of the distance traveled by an aircraft to the altitude lost over the same interval. This ratio gives an estimate for how far a glider is able to travel starting at some altitude, and it is possibly the most important number when evaluating glider performance. The glide ratio can also be expressed as the ratio of Lift to Drag [14]. In general, as the angle of attack increases, the glide ratio increases until a local maximum, and then decreases after the stall condition [5].

From the trade study and historical analysis, it is seen that gliders often prioritize a large aspect ratio, opting for long, thin (short chord length) wings when gliding for long distances. Some early wind tunnel investigations supported this emphasis on aspect ratio, finding that high aspect ratio wings had greater  $C_L$  to  $\alpha$  slopes and greater glide ratios than wings with smaller aspect ratios [2]. A possible explanation for this is that longer wings with higher aspect ratio have less lift reduction from downwash. Downwash would cause an induced drag and an involuntary increase in angle of attack [15].

Another common practice, especially in recreational sailplanes, is the use of winglets, which have been shown to marginally improve the glide ratio when correctly implemented. These winglets should be implemented according to the specific use-case, since some winglets can cause a disproportionate friction and pressure drag at high speeds [16].



**Fig. 8** "Basic study of winglets at two Reynolds numbers showing the effect of the winglets on the lift to drag ratio"[16].

Fig. 8 demonstrates both the effects of winglets and the effect of aspect ratio. Wings 2 and 4 have winglets, and wings 1 and 3 are the same wings except without winglets. Wings 1 and 2 have a greater aspect ratio than wings 3 and 4.



**Fig. 9 Beech Starship (Beech Aircraft Co.) with forward mounted canards.[Air Basic Science]**

Another design consideration is the wing configuration. Attaching additional airfoils such as canards in front of the main wing would trade climb and cruise performance in favor of stability. Canards attached behind the main wing would trade stability in favor of glide ratio. This is not common in sailplanes or gliders, however. Some gliders have an aft-swept wing, which can help achieve higher critical Mach numbers at the cost of spanwise flow and resulting wingtip stall [17]. Critical Mach numbers occur where flow over an airfoil surpasses the speed of sound, and spanwise flow occurs when flow travels along the span of a wing instead of the chord.

Finally, the shape of the airfoil can be optimized to fit the mission requirements. The evidence shows that an airfoil with greater maximum coefficient of lift and a greater lift to angle of attack slope will perform better as a glider. The mistakes made in early aviation show us that a thick airfoil will achieve this goal. Past this, the trade study shows that a cambered airfoil is sometimes used for high-performance applications, with a maximum camber location in the frontward portion of the airfoil. The academic investigation further down in this report will explore the effect of thickness and camber on an airfoil's lift coefficient and lift-slope.

## **F. Conclusion**

The most important metric when evaluating maximum glider performance is glide ratio. A high glide ratio for a given airfoil occurs just before the stall condition, at the maximum lift coefficient. To achieve maximum glide ratio, a large wingspan, a high aspect ratio, and a thick airfoil with some amount of positive camber and winglets should be used. A highly optimized sailplane wing will have a variable airfoil geometry both along the span of the wing and due to control surfaces such as ailerons or flaps. Furthermore, the specific design must adhere to any other design requirements, since handling and ease of use or some other objective is often desirable.

## **G. Connection to Course Content**

The evaluation of glider characteristics is closely related to many ideas in incompressible flow, especially inviscid flow over an airfoil. Since a glider's primary requirement is to remain airborne for as long as possible, or to travel the greatest distance, the glider's wings and their interaction with the air is the focus of any investigation. More specifically, the gliders investigated operate in the subsonic and incompressible regime of air flow. Therefore, an understanding of inviscid flow over airfoils is vital to addressing the problem at hand, in addition to understanding how airfoil and wing characteristics impact the glide ratio of the aircraft.

# **IV. Explanation of Academic-Style Problem**

## **A. Academic-Style Problem Statement**

The goal of the academic problem is to test four different types of airfoils to calculate their lift coefficients. These four airfoils are going to vary in size, they are going to have different cambers and thickness. These different coefficients of lift will create an understanding of what qualities of an airfoil are better or worse for generating lift. These airfoils will be analyzed using different angles of attack to see how that angle would change the amount of lift generated by the airfoil. This problem will use a combined source and vortex panel method to calculate the amounts of lift. A Python program will then be used to calculate different lifts based on different airfoils and angles of attack.

## **B. Assumptions**

- The flow is inviscid, meaning that there is no friction internal to the flow.
- The flow is incompressible, meaning the density of the fluid is constant throughout the flow.
- The flow is irrotational, meaning that there is no vorticity in the flow.
- The flow is steady, meaning the flow does not depend on time.
- The flow is constrained to two dimensions
- There is no flow separation
- The body is impermeable, meaning there is no flow velocity normal to the surface of the body at its surface.
- The flow follows the Kutta condition, meaning the flow smoothly leaves the trailing edge of the body

## C. Methods

The purpose of this research paper is to find the lift coefficients for varying different types of airfoils. The end goal is to see what types of airfoils provide the most lift and what qualities are more favorable to others. The main method that will be used to solve the different lift coefficients is a source and vortex panel method. This method will combine both the source panel method and the vortex panel method to calculate the lift of these different airfoils. The approach for this method is to start by calculating individual point sources and vortexes. The next step will be to put these sources and vortexes in sheets where they will start to represent shapes. These sources and vortex sheets will then be shaped into different shapes to represent these specific airfoils that will be tested. After a final equation for the potential function is obtained, the equation will then be converted into a matrix to solve the missing values. These values will be declared as  $\Lambda$  and  $\Gamma$  and will represent the sources and the strengths of the vortex. After getting the final matrix, a code will take a specific air foil and the matrix to calculate these missing for that airfoil. The code will then take these source and vortex strengths and calculate the coefficient of lift for the airfoil. The code being used is an adaptation for an online public source code developed by a person under the title JoshTheEngineer. Along with the adaptation of his code, his YouTube videos on his channel detailing the methodology of his code were used to help solve this problem.

### 1. Basic Equations for Superposition

To create a simulated airfoil using uniform flow and sources and sinks, the first thing to start with is building potential flow functions. To start, it is important to define the flow is incompressible and mass is neither created nor destroyed. This equation is defined by:

$$\nabla \cdot V = 0 \quad (1)$$

It is also important to declare irrotational flow which is defined by:

$$V = \nabla \phi$$

Combining both the incompressible flow equation and the irrotational equation results in Laplace's equation. Laplace's equation is very popular, and its solutions are known as harmonic functions. This equation is defined as:

$$\nabla \cdot (\nabla \phi) = 0 \quad (2)$$

$$\nabla^2 \phi = 0 \quad (3)$$

Laplace's equation is represented in different coordinate systems by the following equations.

Cartesian coordinates:  $\phi = \phi(x, y, z)$

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$$

Cylindrical coordinates:  $\phi = \phi(r, \theta, z)$

$$\nabla^2 \phi = \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$$

The stream function is then defined by u and v with the following equations:

$$u = \frac{\partial \psi}{\partial y}$$
$$v = -\frac{\partial \psi}{\partial x}$$

The potential function is also defined by u and v with these equations:

$$u = \frac{\partial \phi}{\partial x}$$
$$v = \frac{\partial \phi}{\partial y}$$

## 2. Elementary flow components

The first elementary component that will be defined is uniform flow. When modeling any shape that moves through a free stream velocity, it is important to define the flow with uniform freestream conditions. The flow gets modeled so that the velocity is represented in the X direction as  $V_\infty$

$$\begin{aligned} u &= \frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y} = V_\infty \\ v &= \frac{\partial \phi}{\partial y} = \frac{\partial \psi}{\partial x} = 0 \end{aligned} \quad (4)$$

It is also important to establish a boundary condition in which the flow cannot move through any object in its path. This can be defined by the flow being tangential to the surface and represented by a tangential velocity  $V$ . This velocity will always be perpendicular to the unit normal vector of the surface, which is represented as  $n$ . This wall boundary can be represented by the equation:

$$\nabla \phi \cdot n = 0$$

Or also written as

$$\frac{\partial \phi}{\partial n} = 0$$

To find the potential function of the uniform flow, both the equations from (4) yield:

$$\begin{aligned} \phi &= V_\infty x + f(y) \\ \phi &= \text{const} + g(x) \end{aligned}$$

The  $\phi$  represented in both these equations are the same. This allows you to combine them to one final equation to represent the potential function of a uniform flow:

$$\phi = V_\infty x + \text{const}$$

Which can be simplified as

$$\phi = V_\infty x$$

Doing the same with the stream function, the final stream function for uniform flow is represented as

$$\psi = V_\infty y$$

This gives the final potential and stream functions for uniform flow to be

$$\begin{aligned} \phi &= V_\infty r \cos(\theta) \\ \psi &= V_\infty r \sin(\theta) \end{aligned}$$

Another important type of flow that is commonly used in modeling potential flow is the source flow. The source flow is used to represent an infinite amount of mass entering or leaving a point within a flow field. This is useful to simulate objects within a flow field, specifically airfoils. The strength of the source of is represented by  $\Lambda$ . If  $\Lambda$  is negative, then it represents a sink. A sink is different than a source because a source has an infinite amount of mass flowing out of it, while a sink has infinite mass flowing in it. The source and sink equation are represented in the  $r$  and  $\theta$  directions as the following:

$$\begin{aligned} V_r &= \frac{\Lambda}{2\pi r} = \frac{\partial \phi}{\partial r} \\ V_\theta &= 0 = \frac{\partial \phi}{\partial \theta} \frac{1}{r} \end{aligned} \quad (5)$$

Integrating both the equations from (5) in terms of  $r$  and  $\theta$  respectively we get

$$\begin{aligned} \phi &= \frac{\Lambda}{2\pi} \ln r + f(\theta) \\ \phi &= \text{const} + f(r) \end{aligned}$$

The Greek letter  $\phi$  in both of those equations represent the same value and so combining those equations yield:

$$\phi = \frac{\Lambda}{2\pi} \ln r \quad (6)$$

The stream function of a source or sink is very similar and is represented as the following equations:

$$\begin{aligned} -\frac{\partial \psi}{\partial r} &= V_\theta = 0 \\ \frac{1}{r} \frac{\partial \psi}{\partial \theta} &= V_r = \frac{\Lambda}{2\pi r} \end{aligned}$$

Integrating those equations with respect to  $\theta$  and  $r$  respectively returns the equations:

$$\begin{aligned} \psi &= \frac{\Lambda}{2\pi} \theta + f(r) \\ \psi &= \text{const} + f(\theta) \end{aligned}$$

The Greek letter  $\psi$  is the same in both of the equations, combining those equations leads to:

$$\psi = \frac{\Lambda}{2\pi} \theta$$

Combining both the uniform flow potential and stream function and the source potential and stream functions results in the following equations:

$$\begin{aligned} \phi &= V_\infty r \cos(\theta) + \frac{\Lambda}{2\pi} \ln r \\ \psi &= V_\infty r \sin(\theta) + \frac{\Lambda}{2\pi} \theta \end{aligned}$$

These are a combination of the uniform and source flow within the same free stream. This will be a building block for the panel method.

Circulation is an important measurement to calculate the amount of lift on an airfoil. Circulation is the measurement of the amount of flow that rotates around a solid body and is denoted by the variable  $\Gamma$ . It is represented by the equation:

$$\Gamma = - \oint_l V \cdot d\vec{s} = -V_\theta(2\pi r)$$

This can be rewritten as

$$V(\theta) = -\frac{\Gamma}{2\pi r}$$

Calculating the potential function for the circulation flow is obtained by the following equations

$$\begin{aligned} \frac{\partial \phi}{\partial r} &= V_r = 0 \\ \frac{1}{r} \frac{\partial \phi}{\partial \theta} &= V_\theta = -\frac{\Gamma}{2\pi r} \end{aligned}$$

Integrating both of these equations yields the following equation for the potential function of circulation:

$$\phi = -\frac{\Gamma}{2\pi} \theta$$

Finding the stream function of the circulation of the flow around a solid body is given in a similar matter

$$\begin{aligned} \frac{1}{r} \frac{\partial \psi}{\partial \theta} &= V_r = 0 \\ -\frac{\partial \psi}{\partial r} &= V_\theta = -\frac{\Gamma}{2\pi r} \end{aligned}$$

Integrating this gives the final equation for the stream function of the circulation around a solid

$$\psi = \frac{\Gamma}{2\pi} \ln r$$

### 3. Source and Vortex Sheets

After obtaining the equations for the building blocks to create the panel method, it is good to start with how to obtain one panel. The entire airfoil is made of separate panels that are placed along the perimeter of the airfoil, so starting with one is helpful in understanding the over-all airfoil. The more panels used, the more accurate the airfoil. It is important to understand that these panels are individual source sheets and how to represent them. A simple source is represented by its potential function:

$$\phi = \frac{\Lambda}{2\pi} \ln r$$

Putting multiple of these sources together along a sheet can be represented by this equation:

$$\phi = \int_a^b \frac{\lambda(s)}{2\pi} \ln r(s) ds$$

The potential function is taking the integral along the source sheet from points a to b.  $\lambda(s)$  is represented as the strength of the source sheet as it varies as it moves along the sheet.  $r(s)$  represents the distance from an arbitrary point p to any point along the sheet.

Similarly to source sheets, vortex sheets can be created by taking multiple vortices. A singular point vortex can be written by its potential function:

$$\phi = -\frac{\Gamma}{2\pi} \theta$$

Aligning multiple of these point vortexes along a sheet yields the following equation.

$$\phi = -\int_a^b \frac{\gamma(s)}{2\pi} \theta ds$$

The variables are the same as for source sheets but instead of  $\ln x$  the equation uses  $\theta$  which weights the correct amount of strength for the vortex based on the geometry of the panel.

### 4. Creating Source and Vortex Panel Method

Boundary flow determining the normal velocity constraint

$$V_n = \frac{\partial \phi}{\partial n}$$

Potential function for vortex and source panel

$$\begin{aligned} \phi_s(x, y) &= \int_a^b \frac{\lambda(s)}{2\pi} \ln r(x, y, s) ds \\ \phi_v(x, y) &= -\int_a^b \frac{\gamma(s)}{2\pi} \theta(x, y, s) ds \end{aligned} \tag{7}$$

Taking the equations from (7) the source panel method can be defined in terms of j which is each individual panel as

$$\phi_s = \sum_{j=1}^N \frac{\lambda(s)}{2\pi} \int_j \ln r_j(s_j) ds_j \tag{8}$$

Taking equation (8) the vortex panel method can similarly be represented as

$$\phi_v = -\sum_{j=1}^N \frac{\gamma(s)}{2\pi} \int_j \theta_j(s_j) ds_j \tag{9}$$

The normal and tangential velocity can be represented by using the following equations respectively

$$V_{n_i} = \frac{\partial}{\partial n_i} \phi$$

$$V_{t_i} = \frac{\partial}{\partial t_i} \phi$$

Using those equations, the final normal and tangential velocity equations for source panels are:

$$V_{n_i} = V_{\infty} \cos(\theta) + \sum_{j=1}^N \frac{\lambda_j}{2\pi} \int_j \frac{\partial}{\partial n_j} (\ln r_{ij}) ds_j$$

$$V_{t_i} = V_{\infty} \sin(\theta) + \sum_{j=1}^N \frac{\lambda_j}{2\pi} \int_j \frac{\partial}{\partial t_j} (\ln r_{ij}) ds_j$$

Similarly the final vortex equations for normal and tangential velocity is

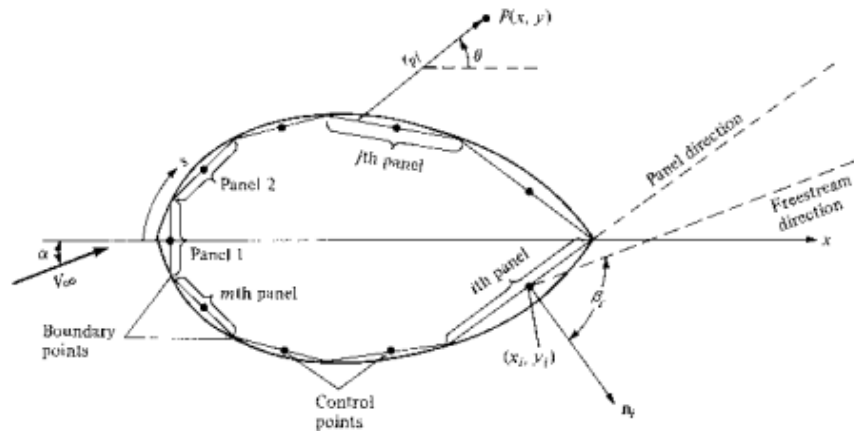
$$V_{n_i} = V_{\infty} \cos(\theta) + \sum_{j=1}^N -\frac{\gamma_j}{2\pi} \int_j \frac{\partial \theta_{ij}}{\partial n_j} ds_j$$

$$V_{t_i} = V_{\infty} \sin(\theta) + \sum_{j=1}^N -\frac{\gamma_j}{2\pi} \int_j \frac{\partial \theta_{ij}}{\partial t_j} ds_j$$

After obtaining these equations, the next step is to create the panel method needed is to combine the source and vortex normal velocity equations. This allows the equation to be a lot more accurate because it uses both the source panel flow and the vortex panel flow to calculate lift and graph streamlines and eliminates the discontinuities present in source panel and vortex alone.

##### 5. Combined Source and Panel Method

When creating the panel method for the combined source and vortex panel it is important to understand the geometry of the problem. This will help identify different types of angles and how they can be used within the equations. This will lead to more simplified equations and a cleaner solution. A good representation of the geometry of the system is in the book Fundamentals of Aerodynamics [5].



**Fig. 10 Panel method geometry for an arbitrary airfoil [5]**

Each point where the panels are connected is the boundary point. These separate each panel from each other. In the middle of each panel is the control point. This is where the vortex and source were centered and analyzed. The more panels and control points, the more accurate the estimate.  $\alpha$  represents the angle of attack, which is the angle of

uniform flow with respect to the horizontal flow.  $\beta$  is the angle of the normal velocity with respect to the angle of attack  $\alpha$ .

The first step in combining these equations is to make the flow streamline by setting the normal velocity equal zero. This is standard for a body moving through a flow and would look like:

$$V_{n_i} = 0$$

To combine both the source panels and the vortex panel, the equations can be added together. This is the same technique used to add the uniform flow and sources or vortexes. The final normal velocity, with the implementation of the new angle definitions, for the combined source and vortex panel method is:

$$V_{n_i} = V_{\infty} \cos(\beta_i) + \sum_{j=1}^N \frac{\lambda_j}{2\pi} \int_j \frac{\partial}{\partial n_j} (\ln r_{ij}) ds_j + \sum_{j=1}^N -\frac{\gamma_j}{2\pi} \int_j \frac{\partial \theta_{ij}}{\partial n_j} ds_j = 0$$

To find the tangential velocity the same method is used. The final tangential velocity for the combined panel source is:

$$V_{t_i} = V_{\infty} \sin(\beta_i) + \sum_{j=1}^N \frac{\lambda_j}{2\pi} \int_j \frac{\partial}{\partial n_j} (\ln r_{ij}) ds_j + \frac{\gamma_n}{2} - \sum_{j=1}^N \int_j \frac{\partial \theta_{ij}}{\partial n_j} ds_j + \frac{\gamma}{2}$$

After obtaining these equations the next steps is to be able to put them in a matrix that our code will be able to solve. The goal is to be able to find the missing  $\Gamma$  and  $\Lambda$  that satisfy the equation. This matrix will then be adapted into code to calculate the amount of lift.

One way to simplify the normal velocity equation is to make the geometric integral into variables that change based on the panels that are being evaluated. This allows the code to look a lot cleaner and be able to be calculated easier. This substitution would look like:

$$\begin{aligned} I_{ij} &= \int_j \frac{\partial}{\partial n_j} (\ln r_{ij}) ds_j \\ K_{ij} &= \int_j \frac{\partial \theta_{ij}}{\partial n_j} ds_j \\ V_{n_i} &= V_{\infty} \cos(\beta_i) + \sum_{j=1}^N \frac{\lambda_j I_{ij}}{2\pi} + \sum_{j=1}^N -\frac{\gamma_j K_{ij}}{2\pi} = 0 \end{aligned}$$

One problem that occurs is when  $J$  is equal to  $I$ . This leads to an error within these equations. To counteract this, an exception must be placed. When  $J = I$ , the geometric integral part of the equation computes the results of when  $J = I$ . This provides a solution for the source and vortex panel method respectively:

$$\begin{aligned} &\sum_{j=1}^N \frac{\lambda_j I_{ij}}{2\pi} \\ &\text{when } j = i \rightarrow \frac{\lambda_i}{2} \\ &\sum_{j=1}^N -\frac{\gamma_j K_{ij}}{2\pi} \\ &\text{when } j = i \rightarrow 0 \end{aligned}$$

Which leads to the final equation of:

$$V_{n_i} = V_{\infty} \cos(\beta_i) + \frac{\lambda_i}{2} + \sum_{j=1, j \neq i}^N \frac{\lambda_j I_{ij}}{2\pi} + \sum_{j=1, j \neq i}^N -\frac{\gamma_j K_{ij}}{2\pi} = 0$$

The velocity of the free stream is then moved to the right to isolate  $\Lambda$  and  $\Gamma$ .  $2\pi$  is multiplied on both sides to simplify even more. These steps then yield the following equation:

$$\pi \lambda_i + \sum_{j=1, j \neq i}^N \lambda_j I_{ij} - \gamma \sum_{j=1, j \neq i}^N -K_{ij} = -V_{\infty} 2\pi \cos(\beta_i)$$

This equation then can be represented as the following matrix

$$\begin{bmatrix} \pi & I_{12} & I_{13} & \dots & -(K_{12} + K_{13} + \dots) \\ I_{21} & \pi & I_{13} & \dots & -(K_{21} + K_{23} + \dots) \\ I_{31} & I_{32} & \pi & \dots & -(K_{31} + K_{32} + \dots) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ I_{n1} & I_{n2} & I_{n3} & \dots & \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \vdots \\ \gamma \end{bmatrix} = \begin{bmatrix} -V_\infty 2\pi \cos(\beta_1) \\ -V_\infty 2\pi \cos(\beta_2) \\ -V_\infty 2\pi \cos(\beta_3) \\ \vdots \\ -V_\infty 2\pi \cos(\beta_n) \end{bmatrix}$$

One problem with this matrix is that there are more variables than equations, so the matrix would be unsolvable. This is where it is important to understand the Kutta-Joukowski theorem. For this situation, the Kutta-Joukowski theorem that works is the following.

$$\begin{aligned} V_{t_n} &= -V_{t_1} \\ V_{t_n} + V_{t_1} &= 0 \end{aligned}$$

This is the representation on the trail edge of the airfoil. It represents the concept that the tangential velocity of the last panel is the same magnitude but opposite then the first airfoil. This equation can be fully written as

$$\begin{aligned} V_{t_1} &= V_\infty \sin(\beta_1) + \sum_{j=2}^N \frac{\lambda_j J_{1j}}{2\pi} + \frac{\gamma_1}{2} - \sum_{j=2}^N \frac{\gamma L_{1j}}{2\pi} \\ V_{t_N} &= V_\infty \sin(\beta_N) + \sum_{j=1}^{N-1} \frac{\lambda_j J_{Nj}}{2\pi} + \frac{\gamma_N}{2} - \sum_{j=1}^N \frac{\gamma L_{Nj}}{2\pi} \end{aligned}$$

Simplifying this equation by grouping similar terms and multiplying each side by  $2\pi$  allows it to match the past equation and put into the matrix above. This equation would be:

$$\sum_{j=2}^N \lambda_j J_{1j} + \sum_{j=1}^{N-1} \lambda_j J_{Nj} - \sum_{j=2}^N \gamma L_{1j} - \sum_{j=1}^N \gamma L_{Nj} + \frac{2\pi}{2} \gamma + \frac{2\pi}{2} \gamma = -V_\infty 2\pi \sin(\beta_1) - V_\infty 2\pi \sin(\beta_N)$$

Which can be furthered simplified into:

$$\sum_{j=1, j \neq 1, j \neq N}^N \lambda_j (J_{1j} + J_{Nj}) + \gamma \sum_{j=1, j \neq 1, j \neq N}^N -(L_{1j} + L_{Nj}) + 2\pi \gamma = -V_\infty 2\pi [\sin(\beta_1) + \sin(\beta_N)]$$

This allows the final matrix to be:

$$\begin{bmatrix} \pi & I_{12} & I_{13} & \dots & -(K_{12} + K_{13}) \\ I_{21} & \pi & I_{13} & \dots & -(K_{21} + K_{23}) \\ I_{31} & I_{32} & \pi & \dots & -(K_{31} + K_{32}) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ J_{31} & (J_{12} + J_{32}) & J_{13} & \dots & -(L_{12} + L_{13} + L_{31} + L_{32}) + 2\pi \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \vdots \\ \gamma \end{bmatrix} = \begin{bmatrix} -V_\infty 2\pi \cos(\beta_1) \\ -V_\infty 2\pi \cos(\beta_2) \\ -V_\infty 2\pi \cos(\beta_3) \\ \vdots \\ -V_\infty 2\pi [\sin(\beta_1) + \sin(\beta_n)] \end{bmatrix}$$

## 6. Implementation of the Code

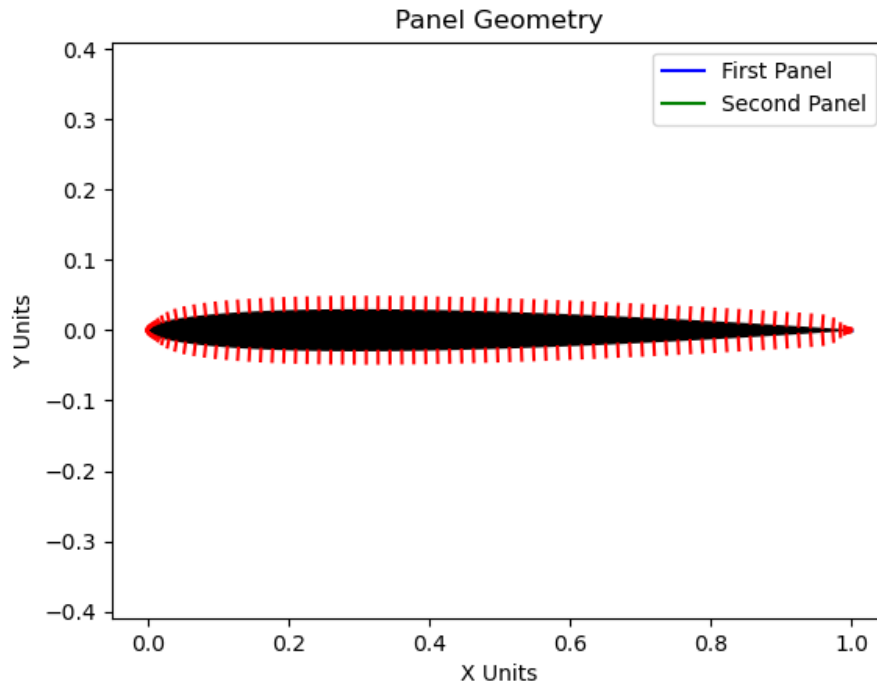
The final matrix that was just solved above will then be solved by a Python script. This Python script was adapted from an online source created by a person that goes under the name of JoshTheEngineer. This code takes a standard NACA airfoil and uploads its dimensions from XFOil. Then a uniform flow is applied at a specified angle of attack. The code will output three different lift coefficients. One from the source vortex panel matrix defined by the above derivations and another from the Kutta-Joukowski lift equation. The final coefficient of lift is what is given by XFOil. Each specific digit for the NACA airfoil has its own meaning. The airfoils that will be defined will all be four-digit NACA numbers. The first digit represents the maximum camber in hundredths of the chord. The second digit is the location of the maximum camber from the leading edge in tenths of the chord. The last two digits are the max thickness in hundredths of the chord. For this problem, four NACA airfoils will be defined. These airfoils will be NACA 0006, NACA 0012, NACA 6406, and NACA 6412.

## D. Results and Discussion

Here each of the individual airfoils will be plotted and analyzed. The code is able to plot six different types of plots. These plots are the airfoil with normal vectors, the geometry boundary points which includes the control points, the airfoil with coefficient of pressure vectors, the pressure coefficient comparison of XFOIL versus the panel method, the airfoil streamlines, and the pressure coefficient contour. After plotting the airfoils each of their coefficients of lift will be compared to their angles of attack.

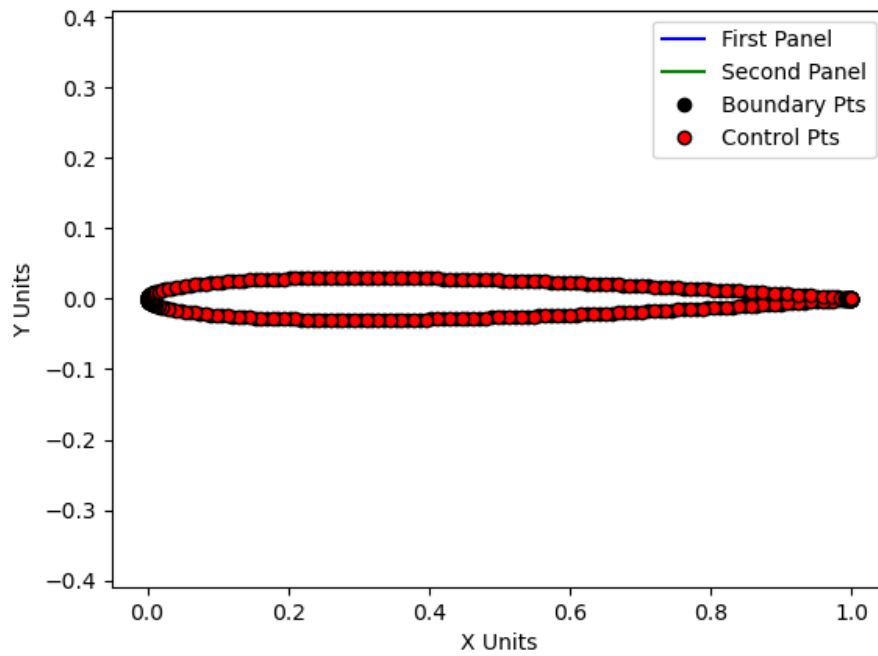
### 1. Airfoil Plots

The first airfoil that will be analyzed is the NACA 0006 airfoil. This is an example of a very thin symmetrical airfoil. This airfoil will give a base of how a basic symmetrical airfoil will act within a uniform flow.



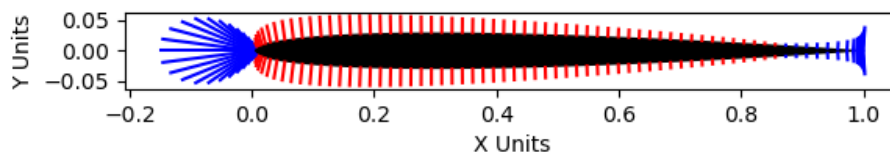
**Fig. 11 NACA 0006 airfoil with normal vectors displayed**

This plot shows the normal vectors of each point on the airfoil. These normal vectors form at the control point of each panel on the panel method airfoil. In the plot, the leading edge and trailing edge have more normal vectors because there are more panels there. This allows the shape to be defined more accurately.



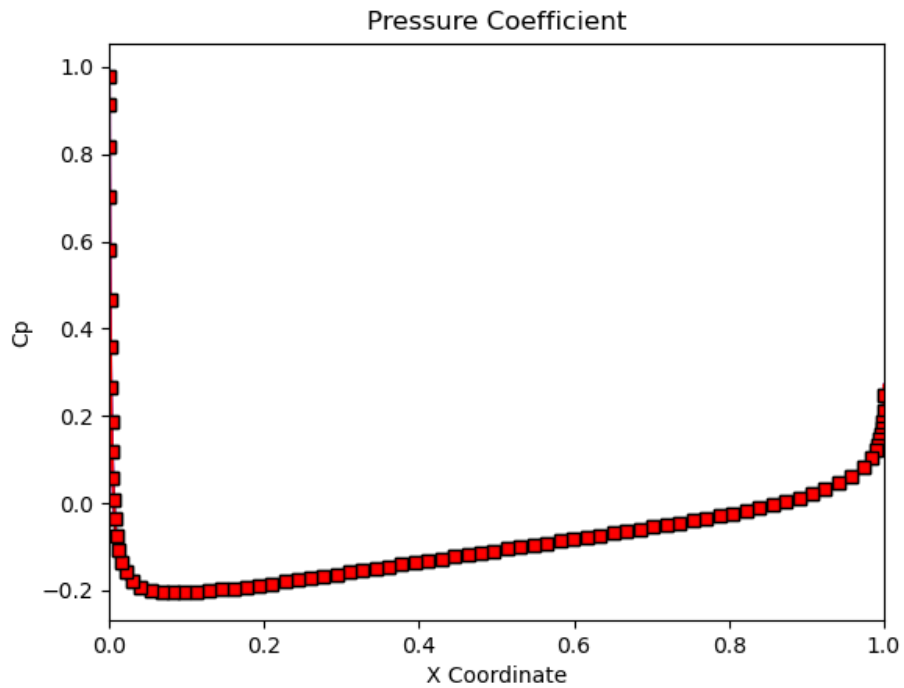
**Fig. 12 NACA 0006 airfoil with boundary and control points**

This plot shows the different boundary and control points on the airfoil. This helps represent the different panels that are making up the shape of the airfoil. Like the normal vectors these control and boundary points are more dense on the leading and trailing edge.



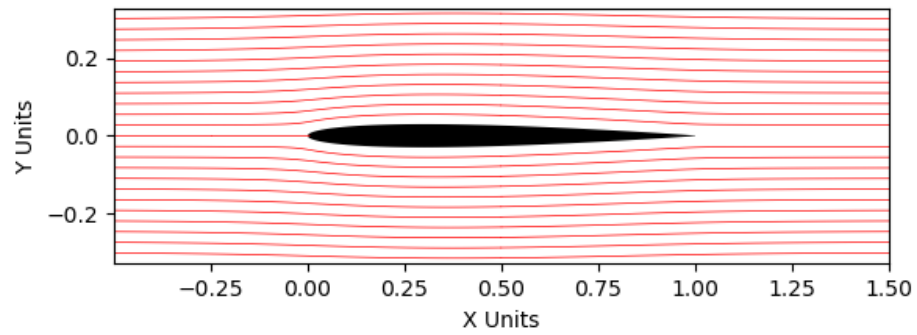
**Fig. 13 NACA 0006 airfoil with coefficient of pressure vectors at each panel**

This plot shows the vectors for the coefficient of pressure on each panel. It displays both the direction and the magnitude of the vectors. The plot shows that the vectors are larger and denser at the front and trailing edges of the airfoil. With the airfoil being symmetrical the plot shows the top and bottom vectors being even because the plot is analyzed at a zero angle of attack.



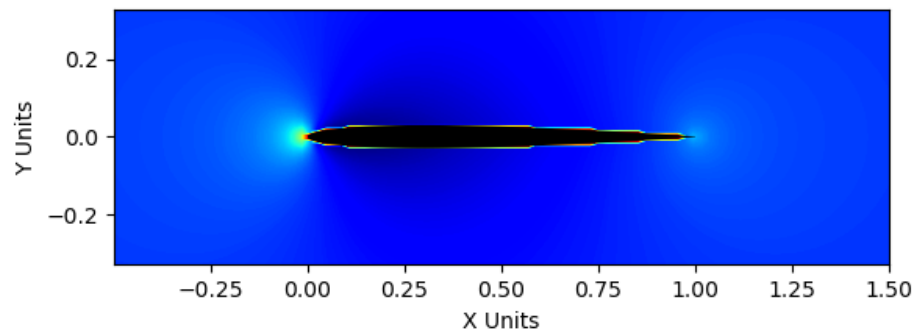
**Fig. 14** NACA 0006 airfoil with pressure coefficients plotted Panel Method vs XFOIL

This plot shows the pressure coefficient as it moves along the airfoil in the x direction. It then compares the pressure coefficients which are the squares to what XFOIL gives, which is the line. The plot shows how close the panel method and XFOIL results are.



**Fig. 15 NACA 0006 airfoil with streamlines**

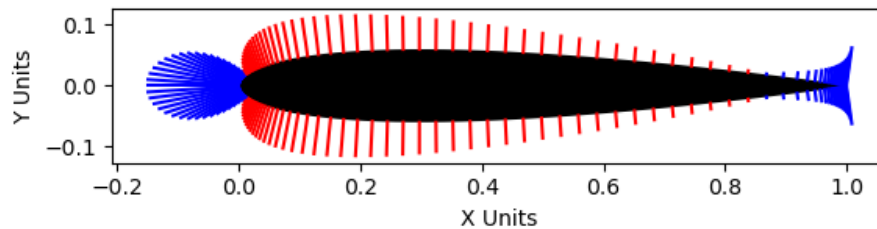
This plot shows the disruption in the streamlines the airfoils causes. The lines show the air being pushed to move around the airfoil.



**Fig. 16 NACA 0006 airfoil with pressure contours**

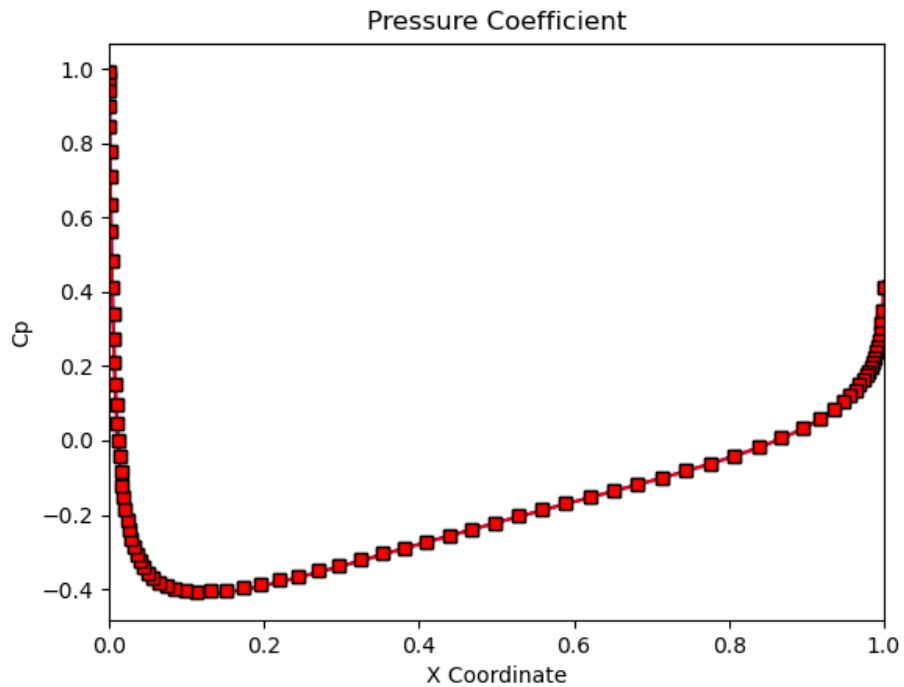
This plot shows the pressure changes within the air due to the airfoil. The plot shows the darker the color the more pressure is in that area. The plot also shows the stagnation points at the front and back of the air foil.

The next airfoil is the NACA 0012 airfoil. This is very similar to the NACA 0006 airfoil, but slightly thicker. This will give a basic understanding on how the thickness of the airfoil can affect the lift.



**Fig. 17 NACA 0012 airfoil with coefficient of pressure vectors at each panel**

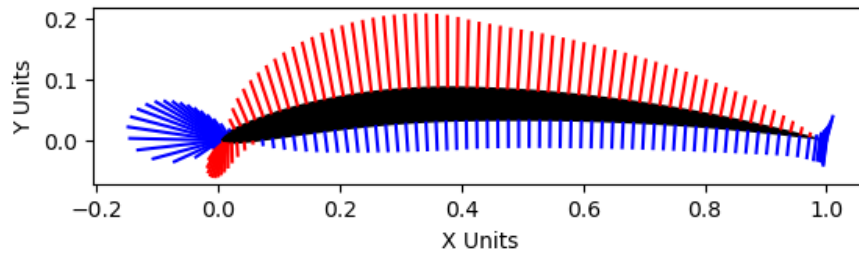
This plot shows the pressure vectors along the airfoil. Compared to the NACA 0006 airfoil these vectors are a lot larger and show that the airfoil disrupts the air a larger amount.



**Fig. 18** NACA 0012 airfoil with pressure coefficients plotted Panel Method vs Xfoil

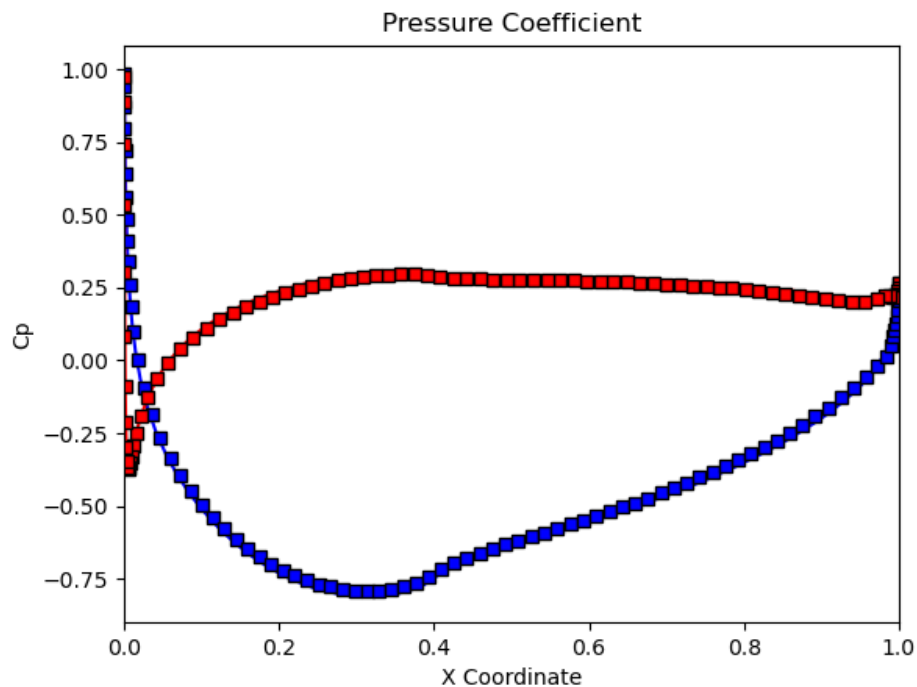
This plot shows the pressure coefficients of this airfoil as it moves in the X direction. Compared to the 0006 airfoil, the pressure coefficients are overall higher.

The third airfoil that will be analyzed is the NACA 6406 airfoil. This airfoil is similar to the NACA 0006 airfoil in thickness. The difference with this airfoil is that it is at a slight camber. The results of the data of this airfoil will help create an understanding of how camber can effect the lift on an airfoil.



**Fig. 19** NACA 6406 airfoil with coefficient of pressure vectors at each panel

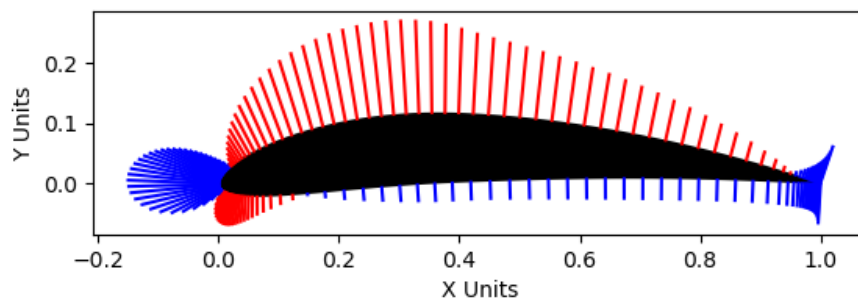
This plot shows the pressure vectors of a cambered airfoil. When adding camber to an airfoil, the plot shows how the lift will be significantly increased. This can be seen in the red vectors in the front of the airfoil, which represents positive lift. The vectors on the top of the airfoil are also much greater than bottom, which shows overall positive lift.



**Fig. 20** NACA 6406 airfoil with pressure coefficients plotted Panel Method vs Xfoil

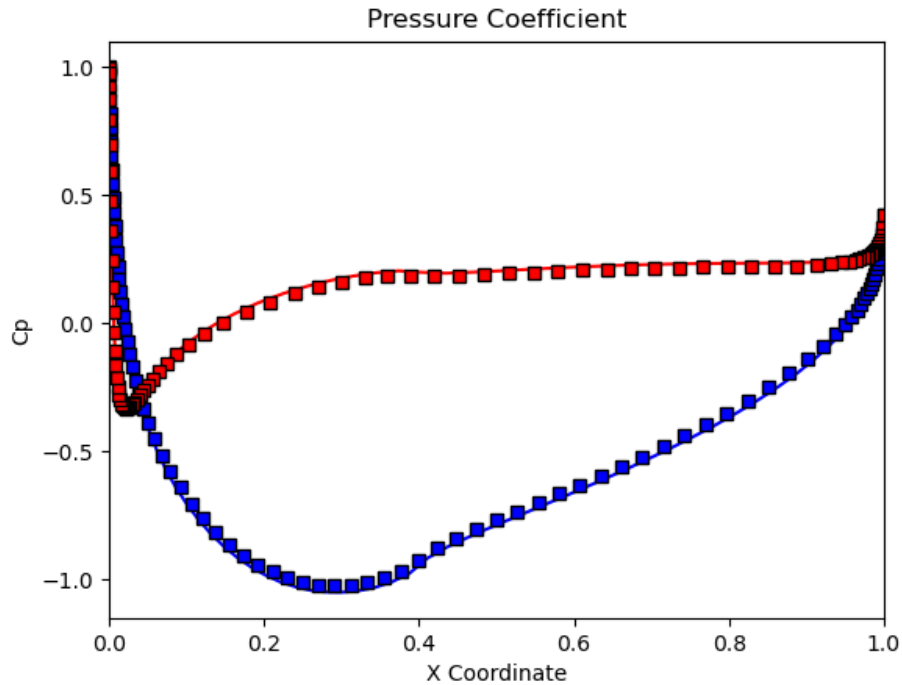
This plot shows the pressure coefficients of the airfoil. With the added camber, the airfoil generates a lot more lift. Leading to a greater change between the positive and negative pressure coefficients as shown in the plot

The final airfoil is the NACA 6412 airfoil. With the same maximum thickness as the NACA 0012 airfoil this airfoil will help show the impact that a thick cambered airfoil has on the lift of an airfoil in a streamline.



**Fig. 21 NACA 6412 airfoil with coefficient of pressure vectors at each panel**

This graph shows the pressure vectors at each panel on this airfoil. Compared to the NACA 6406 the positive vectors are a lot larger which shows an increase in lift.



**Fig. 22 NACA 6412 airfoil with pressure coefficients plotted Panel Method vs Xfoil**

This graph shows the different coefficients of pressure as the X coordinate increases. The positive coefficients are much larger than in the NACA 6406 plot. The difference between the positive and negative coefficients is also the greatest. This is because of the added camber and thickness which produce the most lift.

## 2. Analysis of Coefficients of Lift

With each individual airfoil, the lift coefficients change according to the angles of attack. These lift coefficients can help with understanding how the airfoil will act within a streamline. It can also help to compute the zero angle of the lift angle. Comparing all these airfoils and their individual coefficients of lift will help show which airfoil is most effective and its important properties.

| NACA Airfoil                | NACA 0006   | NACA 0012  | NACA 6406 | NACA 6412 |
|-----------------------------|-------------|------------|-----------|-----------|
| Angle of Attack, $-3^\circ$ | -0.34544    | -0.355848  | 0.3529    | 0.37168   |
| Angle of Attack, $0^\circ$  | -0.00000425 | -0.000003  | 0.70005   | 0.72874   |
| Angle of Attack, $3^\circ$  | 0.345433    | 0.355818   | 1.043705  | 1.0821867 |
| Angle of Attack, $6^\circ$  | 0.6872371   | 0.70785226 | 1.38023   | 1.428242  |

**Table 1 Table showing different airfoils and their coefficients of lift based off of the angle of attack**

This table shows how the angle of attack affects the coefficients of lift of each individual airfoil. It is clear to see that the larger the angle of attack is the greater the coefficient of lift is. One thing that was not analyzed in this problem was stall angle. The stall angle is where the angle of attack is so large the there is too much lift. This causes the airfoil to go vertical and fall. Another factor that affects the coefficient of lift is the size of the airfoil. From both the graph and seen in the plots the greater the airfoil is in size and camber the more lift will be produced. Within the table it shows that the airfoil that produced the greatest amount of lift was the NACA 6412. This airfoil is the thickest and has the greatest camber of the airfoils. The reason that this provides the most lift is because it is able to change the direction of the most amount of air, pushing it downwards. This allows the airfoil to be pushed back upwards generating lift.

## **V. Connection Between the Real-World problem, the Academic-Style problem, and Course Content**

The real world problem explored the design characteristics relevant in glider design, using historical and trade analysis to discover trends and appropriate considerations in the design of a glider for maximum performance. A primary consideration revealed in this investigation was the shape of the airfoil constituting the shape of the glider's wing. While other variables played an important role, this consideration does not afford the designer a straight-forward goal aside from a general idea of thickness and possibly camber. The academic problem examined the expected behavior achieved from particular airfoil selections, directly providing a comparison of a general airfoil shape and examining how adjusting the camber, thickness, or both would affect this expected behavior. The academic problem achieves this by applying inviscid potential flow theory to develop a combined vortex panel and source panel method written in Python and then finding the pressure distribution and/or circulation over any given airfoil. The Real-World problem guides the motivation for the academic problem first to investigate an airfoil, and then to select which airfoils in particular to analyze, affording some meaningful context for the resulting output of the program. Source and vortex panel methods were discussed in detail in lecture, and this adaptation extends the methods from class to combine the two methods along with outputting the results in an easily digestible format.

The primary difference between the Real-World and the Academic-Style problems is the scope. The real world problem takes a broad survey of the important characteristics of a glider, from airfoil to wing shape, aspect ratio and more, and examines how each characteristic impacts the glide ratio metric which underpins the investigation. However, the Real-World problem does not delve into precise details around any of these characteristics. The Academic-Style problem hones in very specifically on the airfoil itself, concerning the investigation with the particular details of an airfoil and making quantitative assessments of physical quantities. Additionally, the Real-World problem makes no assumptions aside from the guiding design goals while the academic problem makes some core assumptions to ensure the mathematical consistency of the model, which is in itself a simplification to avoid prohibitively complicated math. The assumptions made for the Academic-Style problem include inviscid, incompressible, irrotational, steady, 2-dimensional flow, with no flow separation, an impermeable body, and the Kutta condition. The effect these assumptions have on the physical model is that the model does not consider or predict viscous drag due to the inviscid assumption; cannot model stall or flow separation and therefore cannot predict the maximum lift coefficient for an airfoil; and does not account for time-dependent phenomena such as vortex shedding. As a consequence, the choice of results from the model and how these results are interpreted must be limited to account for these losses in physical resolution. For example, while the real world problem would hope to use a comparison of maximum lift coefficients to evaluate performance, the model will not predict this value, and therefore it must be measured from experimentation. The model can be used, however, to predict the lift coefficient at a given angle of attack as long as this lift coefficient is below the maximum lift coefficient before stalling.

## **VI. Team Management**

### **A. Project Tasks and Schedule**

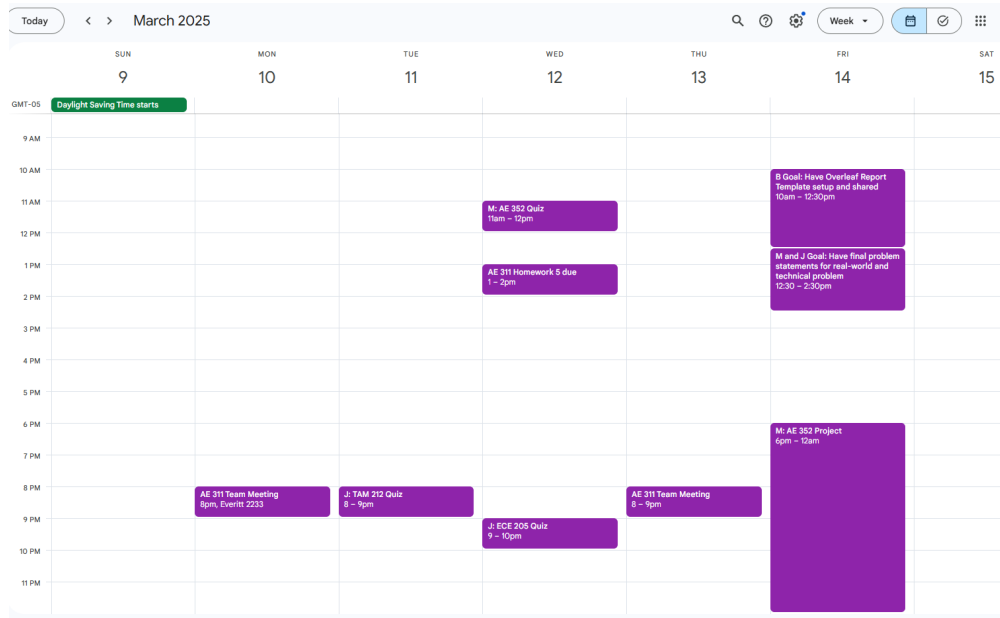
The Real-World problem lead was tasked with researching the history of gliders and how they became more complex as the human understanding of aerodynamics increased. This involved not only the aircraft but also the methods used to study them. Then, the Real-World problem lead had to hone in on information directly relevant to the course content, building a bridge between the broader scope of aerodynamics and the material that is learned in class.

The Technical Lead was tasked with applying the concepts learned in lecture to create an interesting academic-style problem relating to content that has been covered in lecture. This meant taking elements of the Real-World problem adding simplifications or assumptions to create a situation that could reasonably simulated with material learned in class. From there, it was a matter of ensuring the problem was difficult enough to meet the expectations of the project and that the solution was correct.

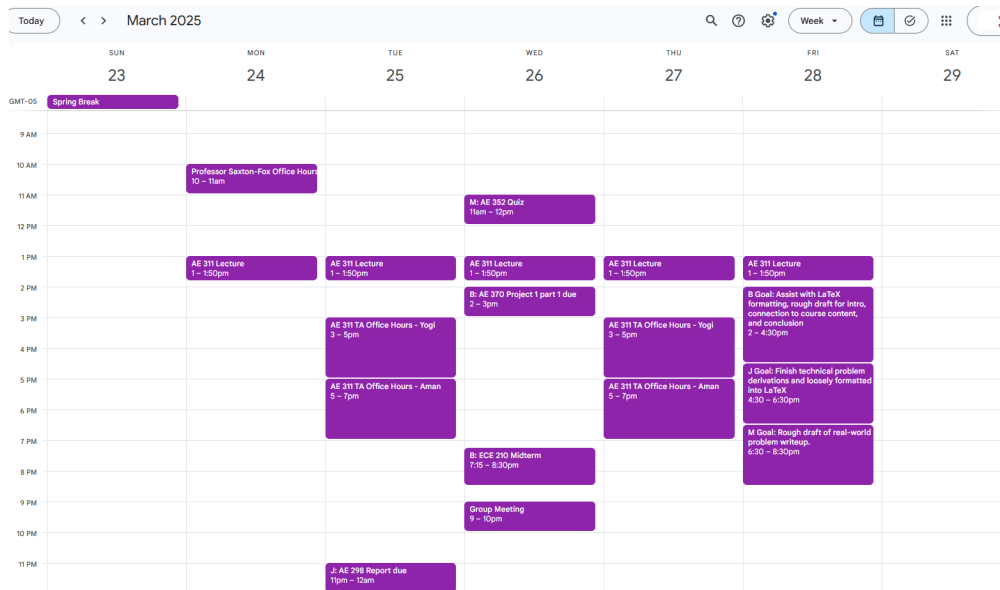
The Team Manager's role was tasked with planning meeting times, organizing the structure of the project, and resolving disputes or any other difficulties that may have obstructed the team in their pursuit of the goal.

The project schedule was consolidated into a Google Calendar containing weekly goals, important AE 311-related events, and key events pertaining to each group member outside of AE 311. Important AE 311 events included lecture meeting time, professor office hours, TA office hours, and deadlines. Important events outside AE 311 included quizzes, tests, project deadlines, and extracurricular events. The goal of this calendar was to make planning for group meetings and work sessions easier. There was less friction to planning since any group member could check when the others were

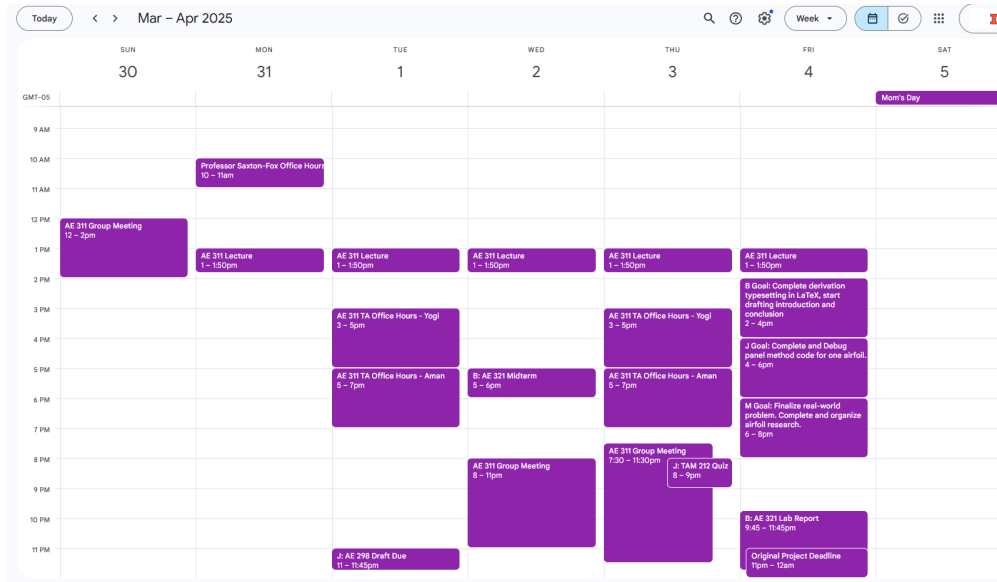
free. The Google Calendar is shown here:



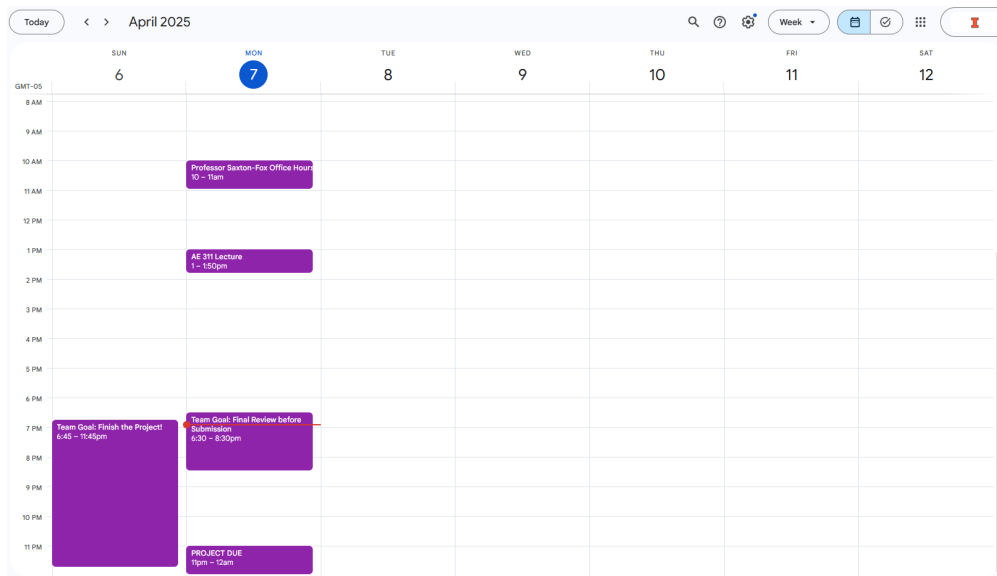
**Fig. 23 Team Schedule for the Week of March 9th**



**Fig. 24 Team Schedule for the Week of March 23rd**



**Fig. 25 Team Schedule for the Week of March 30th**



**Fig. 26 Team Schedule for the Week of April 6th**

The weekly goals were an easy way for the group to assign reasonable tasks specific to each group member to ensure that progress toward the end result was made each week. Each member's goal was discussed and decided on by the group. This practice was helpful with synchronizing the work of different members, especially the Real-World and Technical leads. Their work often relied on the other's to make sure that their problem statements remained connected.

Each meeting, notes were taken to document progress, adjust goals if needed, and assign action items to team members within their overarching tasks. This worked as a way to lend extra structure to the project timeline. Breaking large tasks into smaller chunks at each team meeting also made the project less intimidating. Below is an example of how notes were recorded at the team meetings:

## Apr 2, 2025 | 📅 AE 311 Group Meeting

Attendees: Max Lutter, Jackson Rees, Ben McDonald

### Notes

- Project Deadline got pushed back to Monday, April 7th
- Take inventory of where we are right now
  - Jackson's panel method code is up and running
  - Max's research is almost done, sections of the writeup have been drafted
  - All three members are now able to focus almost solely on finishing up the project

### Action items

- ☐ Ben:
  - ☐ Start tedious formatting tasks (citations, equation derivation...)
  - ☐ Begin drafting communication, challenges, and lessons sections
- ☐ Jackson:
  - ☐ Map out equation derivation for report
  - ☐ Look into importing and testing other airfoils in panel method
- ☐ Max:
  - ☐ Continue researching, hone in on real-world problem statement
  - ☐ Research relevant glider airfoils for Jackson to test
  - ☐ Provide Airfoil for Jackson
  - ☐ Thursday → finish research
- ☐ Group:
  - ☐ Hoping to complete the project by Sunday at the latest
    - ☐ Finishing up on Monday would interfere too much with other schoolwork
  - ☐ Rough draft of report completed for Sunday morning

**Fig. 27 Notes Document From the Team's Meeting on April 2nd**

## B. Communication

Communication was integral for success in this project. The team continued to use an iMessage group chat that was made during Project 1 as the primary form of communication. In this group chat, the team would discuss meeting times, day-to-day progress related to the project, and any quick questions or points of concern that could be resolved over text. Group video calls through FaceTime were also important for members to get on the same page and talk about issues when getting together was not realistic or efficient. Communication between the Real-World and Technical Leads was most important to ensure that the Real-World and Academic problems were aligned in their content, approach, and end goals.

Communication throughout the project remained respectful and constructive. The Team Manager would occasionally act as a liaison between the other members to make sure that expectations from one problem lead to another were clear and attainable.

Communication with the professor and teaching assistants (TAs) was used throughout the project to make sure that the project aligned with course expectations. The Technical lead went into the professor's office hours to make sure that coding a panel method was feasible. The Technical lead also visited TA office hours to verify components of the solution as the project went on.

## C. Challenges and Lessons Learned

### 1. Lessons

Although last project was a big success, we still had lessons to take from it. The most prominent lesson we learned was the amount of time it takes to format equations into LaTeX. Last project, it was something that was saved until the very end, causing us to cut our submission incredibly close to the deadline. As a result, I urged the Technical lead to

map out the derivations early so I could start working on them much earlier in the project window. More generally, all of the formatting and "finishing touches" (the references section, nomenclature section, inserting and formatting images, etc.) take much longer than anticipated. Similarly, I started these portions earlier in the project window to reduce stress near the end. The group has also become more familiar with Overleaf's layout and LaTeX formatting, meaning that we have not gotten hung up on layout or syntax in Overleaf nearly as much as the first project.

## *2. Challenges*

The greatest challenge throughout the project was scheduling times to meet together. Spring Break was in the middle of the project window; some group members left early for Spring Break, and some members returned late. With the end of Spring Break came a rush to make up work, as the team members decided not to complete any work over Spring Break. This led to a slow start on the project coming back from break, further increasing stress and tension surrounding the project.

It also took the Real-World and Technical leads a longer time to settle on a problem statement than last project. They got caught in a loop of sorts where each member expected more information from the other before continuing without properly communicating their needs. As a result, I sought to facilitate the discussion between the two by explicitly asking each group member what they required from the other. That way, the group could move closer to a common problem with greater understanding.

Differences in lifestyles also served as an obstacle to overcome. Two members preferred to get most of their work done late at night, while the third member preferred to sleep early. This added an extra constraint on meeting times that I had to navigate so that we found blocks of time where everyone could meet without encroaching on anyone's personal preferences and habits. Thorough communication through the group chat was the best way to work around these issues before they arose.

## **VII. Conclusion**

The Real-World problem explored the many parameters that must be considered when designing a glider. By pulling from historical developments in the field and modern-day trade studies, a comprehensive overview of the effect of airfoil design, wings, wing tips, and aspect ratio contribute to a glider which can achieve the longest flight. The Academic-Style problem shifted the design focus specifically on the choice of airfoil. 4 different airfoils were analyzed: a thin symmetric airfoil, thick symmetric airfoil, thin cambered airfoil, and a thick cambered airfoil. The goal was to see how the airfoil characteristics and angle of attack affected lift and pressure distribution. This was accomplished through coding an airfoil panel method in Python which took NACA data to calculate lift coefficients and pressure distributions for each airfoil. The Academic-Style problem related to the Real-World problem by tailoring calculations and results to examine airfoil performance in usual glider environments. These problems both connect to key topics in the course such as potential flow, panel methods, lift, and airfoils. Gliders heavily rely on an understanding of fluid mechanics to be able to generate lift at relatively low speeds and low angles of attack. The panel method was also reviewed across multiple lectures as a way to leverage potential flow to calculate accurate lift values within a meaningful range of angles of attack. Hence, this report introduced a problem which ties heavily to concepts throughout incompressible flow and expands upon a solution method covered in lecture.

The Team Manager placed an emphasis on efficiency in team meetings. This project window it was more difficult to find times to meet with clashing schedules and Spring Break in the middle. As a result, the Team Manager communicated frequently both in the group chat and individually with each member to ensure that the team was prepared to work effectively at each meeting. A group calendar was shared to the team to familiarize the team with each other's schedules and availability as well as key AE 311 events like office hours. Additionally, the Team Manager made sure that expectations were communicated clearly by taking notes at each in-person meeting and working with the team to set weekly goals for each group member throughout the project.

## **Appendix**

### **A. Use of Artificial Intelligence**

Artificial Intelligence (AI) was used throughout the project by the Team Leader to facilitate LaTeX formatting in the report. For example, AI was prompted to provide a matrix layout for LaTeX, which was copied and pasted into the report and adapted to fit the equation derivations.

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